DOLPHINBENCH · PERSONA RILEY · RUN 2026-11-05

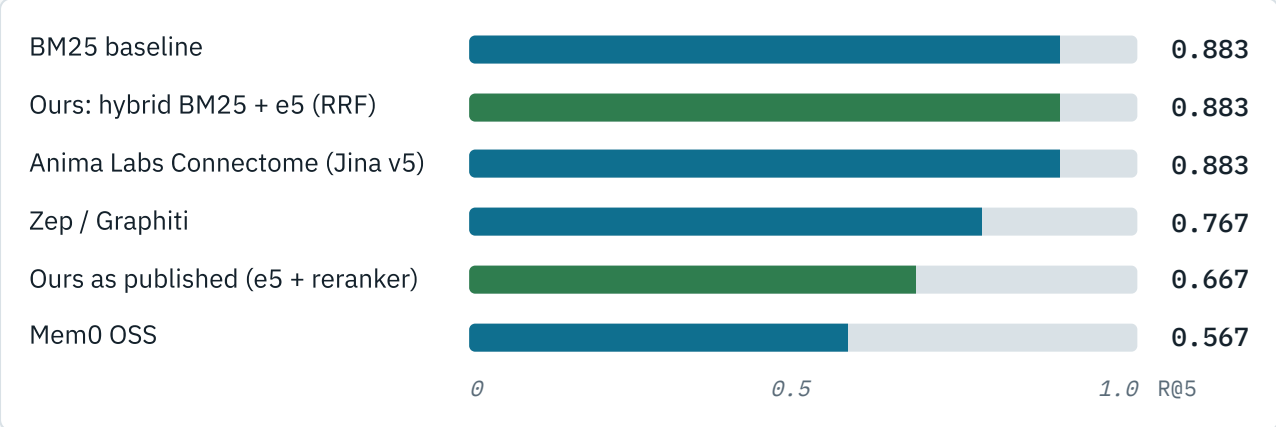
Retrieval layer on DolphinBench: Mem0, Zep, Anima and our search

DolphinBench grades the action an agent takes after ~500K tokens of history. Before running the full action suite, we measured the layer every action depends on: does the memory return the history sessions that the benchmark's own `facts.yaml` marks as the source for each test, and how fast.

Method

- Gold: for each test, `load_bearing_facts` → `source_session_ids` from the released annotations.
- Query: the test request text, unchanged. Output: ranked history sessions. Recall@k = share of gold sessions in the top k.
- Same corpus for every system: all gold sessions plus seeded random distractors, each session prefixed with its date. 50 random tests (seed 7).
- Mem0 OSS 2.2.1 with BM25 and spaCy enabled, Graphiti 0.31.2: both on gpt-4.1-mini and text-embedding-3-small.

All memory systems: 20 tests, 200 sessions

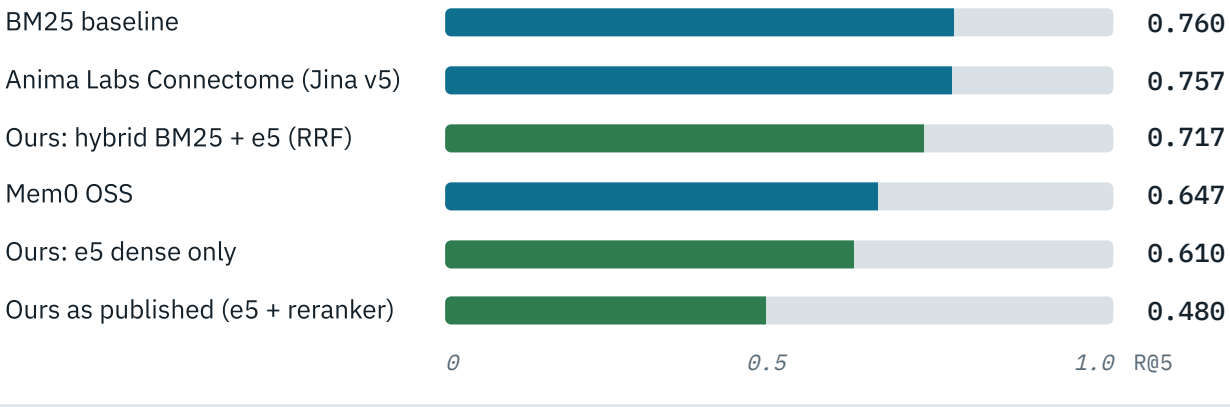


SYSTEM	RECALL@5	RECALL@10	ANY GOLD IN TOP 5	ALL G
Mem0 OSS 2.2.1	0.567	0.733	0.70	
Anima Labs Connectome (Jina v5, GPU)	0.883	0.908	1.00	
Zep / Graphiti 0.31.2	0.767	0.833	0.90	

SYSTEM	RECALL@5	RECALL@10	ANY GOLD IN TOP 5	ALL G
Ours: hybrid BM25 + e5 (RRF)	0.883	0.883	1.00	
Ours: e5 dense only	0.792	0.842	0.95	
Ours as published (e5 + reranker)	0.667	0.758	0.85	
BM25 baseline	0.883	0.925	0.95	

Misses in the top 5 on this slice: Mem0 6 of 20 tests, Graphiti 2, Anima 0, our hybrid 0, BM25 1. Graphiti ran on embedded Kuzu under Linux (WSL); on native Windows the same Kuzu build stalled.

50 tests, 1,000 sessions (~120K tokens)



SYSTEM	RECALL@5	RECALL@10	ANY GOLD IN TOP 5	ALL G
Mem0 OSS 2.2.1	0.647	0.743	0.72	
Anima Labs Connectome (Jina v5, GPU)	0.757	0.833	0.84	
Graphiti 0.31.2 (Kuzu)	not run at this size: sequential ingest at ~9 s/session is ab			
Ours: hybrid BM25 + e5 (RRF)	0.717	0.843	0.80	
Ours: e5 dense only	0.610	0.697	0.68	
Ours as published (e5 + reranker)	0.480	0.577	0.54	
BM25 baseline	0.760	0.793	0.84	

Full Riley history: 50 tests, 5,128 sessions (~500K tokens)

SYSTEM	RECALL@5	RECALL@10	ANY GOLD IN TOP 5	P50 S
Ours: hybrid BM25 + e5 (RRF)	0.540	0.650	0.60	0.43
Ours: e5 dense only	0.400	0.470	0.46	0.27
Ours as published (e5 + reranker)	0.220	0.360	0.28	3.19
BM25 baseline	0.590	0.643	0.66	0.07

Mem0 was not run on the full history: 1,000 sessions took 22 minutes of ingest, 5,128 would take about two hours.

What we learned

- **Lexical signal is strong on this set.** Tests name projects, channels and people, so plain BM25 beats every dense setup. A hybrid closes most of the gap for us.
- **Our cross-encoder reranker hurts** by 12 to 18 points of Recall@5. The benchmark found a regression our own eval missed.
- **The shared failure is temporal.** 7 of 50 tests are missed by both Mem0 and our hybrid. In these, one topic recurs dozens of times across years and the test points to one meeting by date ("the April 21, 2023 staff review"). Similarity search finds the topic but not the moment.
- **Mem0 and our hybrid fail on different tests** (3 found only by Mem0, 7 only by the hybrid), so the two mechanisms are complementary.

Limits

- This is source-session recall, not the DolphinBench pass rate. Finding the source is necessary for the action, not sufficient.
- One persona, 50 tests: one test is two points, so gaps under five points are noise.
- The hybrid was added after seeing the 1,000-session slice. It held on the full history, but weights were not tuned on held-out tests.
- Mem0 recall is mapped through the `session_id` metadata attached at add time; merged memories keep the first session's tag, which can undercount Mem0.
- Graphiti used the embedded Kuzu backend (deprecated upstream) under Linux, not Neo4j or FalkorDB, and was measured on the 200-session slice only. Its ingest is sequential by design, Mem0's ran with 8 workers, so ingest times are not like for like.
- Anima was measured through its Connectome embeddings index (authentic server, Jina v5 omni small on a local GPU); its agent-written autobiography layer needs a full agent run and was not exercised.

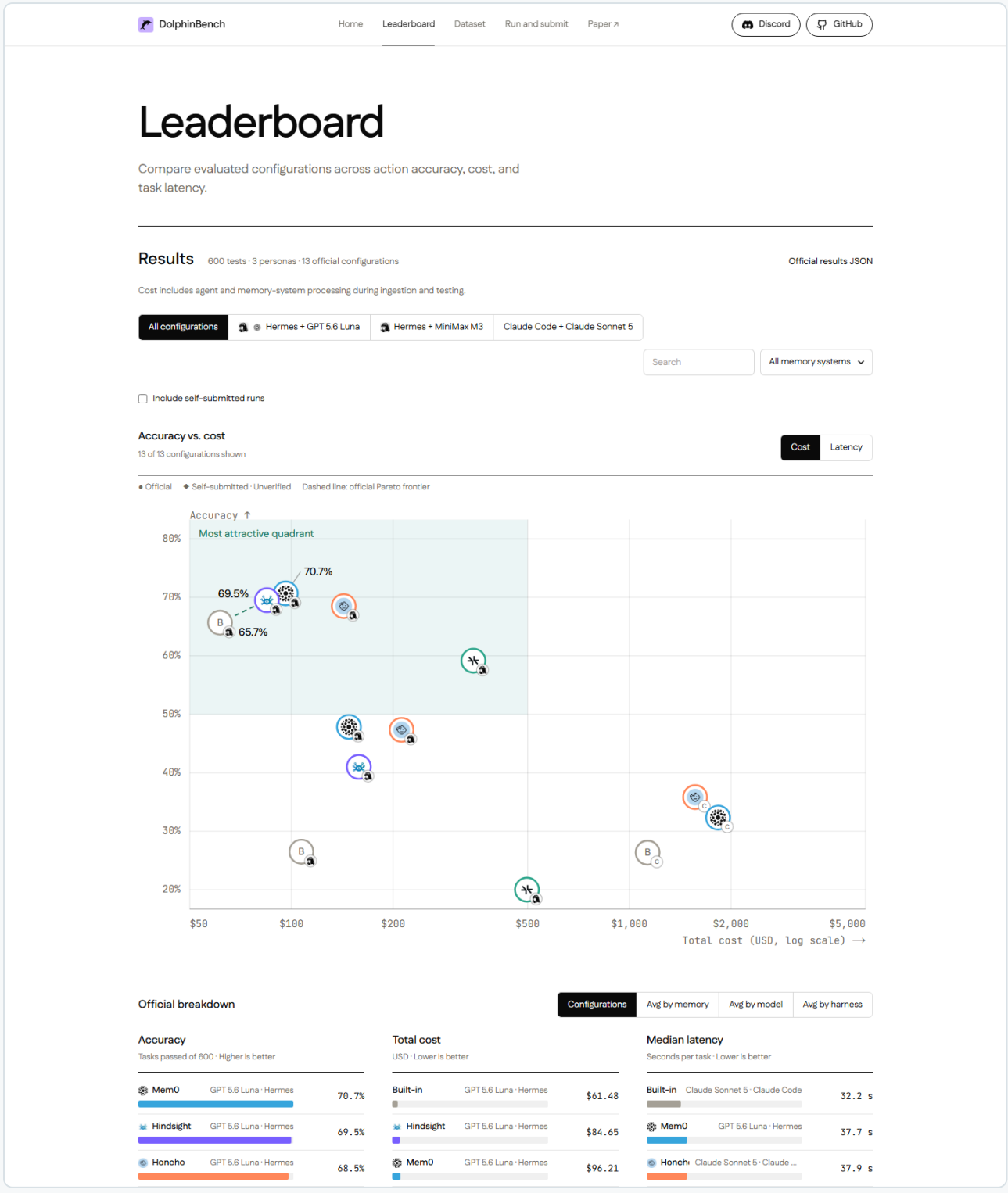
- The 200-session slice has 20 tests: one test is five points.

Official DolphinBench board, for context

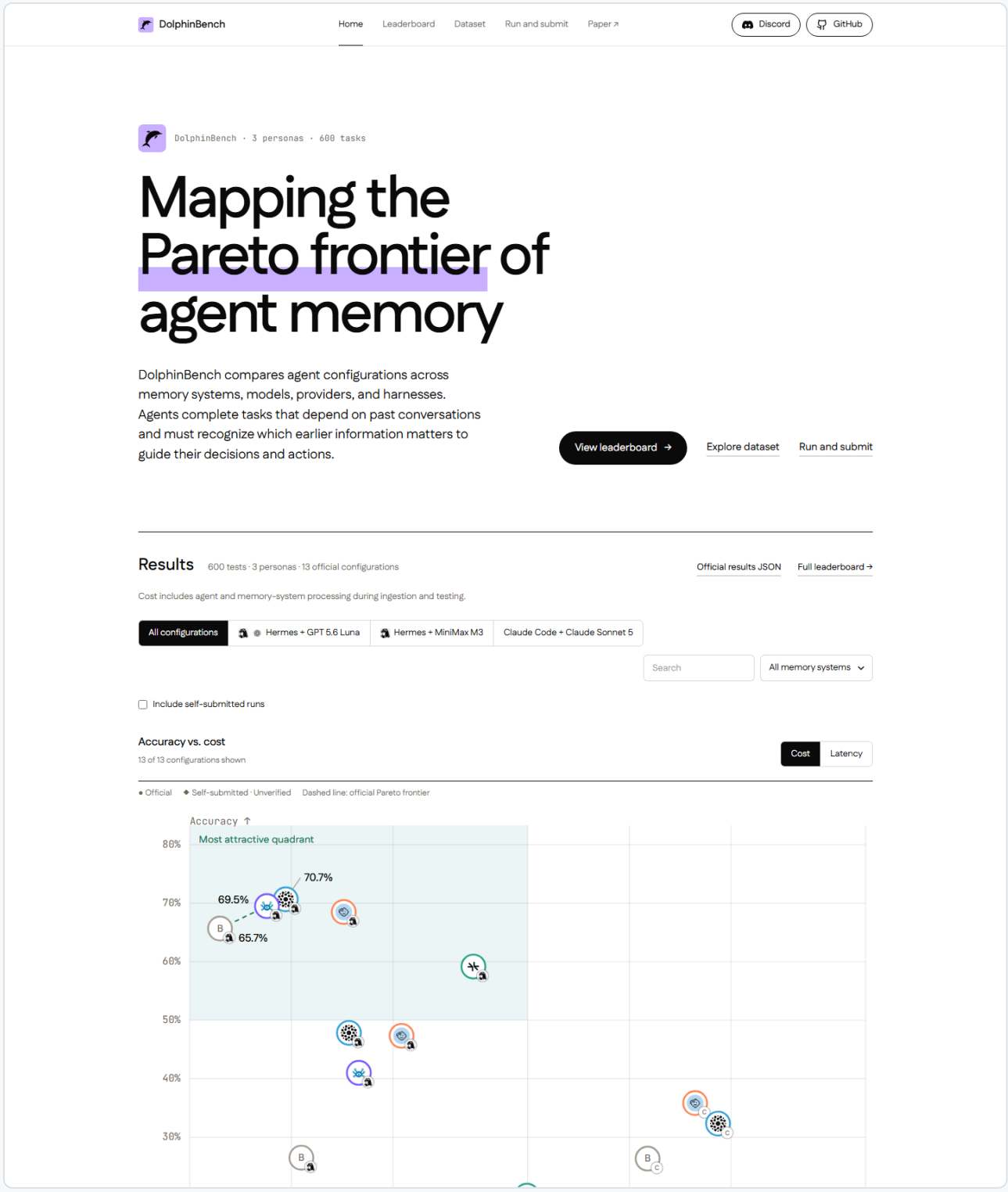
HARNESS · MODEL	MEMORY	PASSED / 600	TOTAL $	MEMORY $	MEDIAN S
Hermes · GPT-5.6 Luna	Built-In	394 (65.7%)	61	0	44.4
Hermes · GPT-5.6 Luna	Mem0	424 (70.7%)	96	35	37.7
Hermes · GPT-5.6 Luna	Hindsight	417 (69.5%)	85	27	55.3
Hermes · GPT-5.6 Luna	Honcho	411 (68.5%)	143	46	44.7
Hermes · MiniMax M3	Built-In	159 (26.5%)	107	0	51.4
Hermes · MiniMax M3	Mem0	287 (47.8%)	148	36	47.0

Source: `results/` in github.com/mem0ai/dolphinbench (Apache-2.0). Memory adds 5 points on the strong model and 21 on the weaker one.

DolphinBench interface: leaderboard at dolphinbench.ai (accuracy vs cost)



DolphinBench interface: home page



Reproduce: `prep.py riley 50 1000`, then `run_arm.py <slice>` `bm25 | sgm | mem0 | graphiti`. Raw per-test rankings kept for every system.